

NORMALIZED FIXED-STRIDE DESCENT: DIVERGENCE-FREE OPTIMIZATION AND A LEARNING-RATE CALIBRATION ERROR IN A FROZEN READOUT INSTRUMENT

A P R E P R I N T

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2 seeds × 2 feature sets, every registered inequality

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ABSTRACT

This study originated in an open-ended design brief. The resulting method — normalized fixed-stride descent (NFSD): a unit-norm momentum direction, a fixed step length, and each candidate step accepted or rejected on the current minibatch — is benchmarked here against the research program's frozen training instrument on two feature sets. Three results, all replicated on a second seed. (1) NFSD **never triggered the divergence tripwire**: zero divergence-tripwire events across four step sizes spanning three orders of magnitude, 2 seeds × 2 feature sets, in a sweep where the reference Adam configuration at $\eta=0.1$ voided three times out of four and, in the fourth cell, diverged to 8.6 bits per character (bpc) *without tripping the tripwire at all*. (2) NFSD exposes a calibration error in the instrument: the reference logistic readout's fixed learning rate (Adam, $1e-3$) is off-optimum in a conditioning-dependent way — by 0.017–0.018 bits on the ill-conditioned feature set, 0.013–0.014 on the well-conditioned one — so every cross-feature-set comparison of levels the program publishes now carries a ~ 0.02 -bit error bar. (3) The step-acceptance test is dead weight: probing each candidate step before accepting it is **strictly harmful** at large stride, 4/4 cells. The safety comes from normalization, not from step probing.

Keywords stochastic optimization · normalized gradient methods · fixed step size · momentum · Adam · learning-rate calibration · pre-registration

§1 Motivation

This study originated in an open-ended design brief. Underneath the brief sits a question the research program needed answered: every representation the program scores passes through one frozen instrument — a logistic readout trained by Adam [1] at learning rate $1e-3$. That instrument's validation loss is treated as *the* value of a representation. Is its learning rate well set on every feature set it is asked to score? And what does it cost to train with an optimizer whose step length is bounded by construction?

Divergence has a precise operational meaning here. The program applies a *divergence tripwire*: a fit-invalidating rule that voids any run whose validation loss blows up during training. In this study that rule is the primary measurement.

§2 Method

Normalized fixed-stride descent (NFSD) is an optimizer for the program's reference logistic readout (27-way softmax, batch 8192, gradient clip 1.0, exponentially weighted moving-average momentum [2] $\beta=0.9$, validation early stopping). Take the momentum direction, normalize it to unit norm, and step a *fixed stride* s . Before a step is accepted, the candidate is evaluated on the current minibatch; if it fails to reduce the minibatch loss, the stride is halved and the candidate re-evaluated (up to 3 backtracks, after which the step is refused); 25 consecutive accepted steps without backtracking grow the stride. The full machine is therefore NFSD with minibatch step acceptance.

```
# unit-norm direction: fixed step, independent of gradient magnitude
d = m / ||m||
# candidate step evaluated on the current minibatch before acceptance
W ← W - s · d
```

Baselines: the reference Adam configuration (the instrument's own optimizer class, verbatim) and momentum SGD, all on identical features and identical seeded batch order, over the program's standard text8 splits [4]. Two ablations isolate the two components: noprobe (probe-free NFSD — normalized fixed stride only, with no per-parameter state and no step acceptance test) and nonorm (unnormalized probed descent — step acceptance without normalization). Two feature sets are used. The **well-conditioned** set is the program's reference sparse-tanh echo state network (ESN) features [3] at their best operating point (spectral radius $\rho=0.8$); the **ill-conditioned** set is the addressed-memory ESN's *linear-activation* variant at $\rho=0.95$ (permutation recurrence with Hadamard-coded inputs) — the worst-conditioned features in the program's inventory (see whitepaper 01 for that construction). Step size was swept across four step sizes spanning three orders of magnitude, $\eta \in \{1e-4, 1e-3, 1e-2, 1e-1\}$: 32 fits per seed.

§3 Pre-registered predictions

Six ordinal predictions (G1–G6) and four quantitative bands were registered before the run. The program registers *named alternatives* — each prediction is recorded together with the reading that would follow from its failure — and that discipline matters here, because the principal finding arrived through the misses.

- **G1**, the headline: NFSD voids in 0/8 sweep cells while at least one baseline voids. (Hit.)
- **G2**, a null expected to hold: NFSD should *not* beat the instrument's operating point by more than 0.01 bits. Named alternative, written in advance: if it does, the instrument's learning rate is off-optimum — provisional, replication automatically queued. (The alternative fired on the ill-conditioned set.)

- **G4**: the optimizer ordering Adam < NFSD < SGD transfers across feature sets. (Miss, informatively: on the ill-conditioned set, optimizer choice at each method's best η stopped mattering at all.)
- **G5**: mechanism claims — where NFSD's step refusals fire and where its worst cell sits. (One miss: full refusals *never* fired; the mechanism operated one level up, in stride-halving backtracks.)
- **G6**: step acceptance should earn its keep against noprobe. Named alternative: step probing is redundant, and the safety comes from normalization rather than from step probing. (The alternative fired, strongly.)

The rerun registration (seed 1) then promoted the surprises to registered claims R1–R4 with a promotion rule stated before running: R1 (ill-conditioned set: NFSD beats the instrument's learning rate by >0.01) *and* R2 (well-conditioned set: noprobe@0.1 beats the instrument's learning rate) both holding on the fresh seed would promote the finding — that is, enter it in the program's confirmed-findings record; any split result would be recorded without promotion.

§4 Results

Table 1. Seed 0, logistic validation bpc (lower is better). VOID = the divergence tripwire fired. Reference instrument setting = adam@1e-3 (bold row).

η	well-cond. Adam	well-cond. SGD	well-cond. NFSD	ill-cond. Adam	ill-cond. SGD	ill-cond. NFSD
1e-4	2.8319	3.7530	3.1719	3.1650	3.4968	3.1979
1e-3	2.6519	3.2842	2.8184	3.1830	3.2267	3.1647
1e-2	2.6577	3.0634	2.6502	3.3763	3.1704	3.1907
1e-1	VOID	2.8064	2.6558	VOID	3.1693	3.3473

Ablations at $\eta = 1e-3 / 1e-1$ — well-conditioned set: noprobe 3.0165 / **2.6384**, nonorm 3.1516 / 2.6937; ill-conditioned set: noprobe 3.1749 / 3.2441, nonorm 3.1828 / 3.2035. As a consistency check, the verbatim adam@1e-3 arm reproduced the archived instrument values (2.6519 well-conditioned, 3.1830 ill-conditioned) to all four decimals.

Reading the grid: NFSD voided nowhere and never exceeded 3.3473 bpc, whereas Adam at $\eta=0.1$ tripped the divergence tripwire on *both* feature sets. On the ill-conditioned set, NFSD@1e-3 (3.1647) beat the reference instrument setting (3.1830) by 0.0183 — firing G2's named alternative — while tuned Adam at 1e-4 reached 3.1650, so NFSD *ties tuned Adam* to within 0.0003; the discrepancy is with the instrument's fixed learning rate, not with Adam's attainable optimum. On the well-conditioned set, noprobe@0.1 (2.6384) was the best cell in the entire grid, 0.0135 below the reference instrument setting. Step acceptance was worse than useless at large stride: full NFSD lost to its own noprobe ablation by +0.0174 (well-conditioned) and +0.1032 (ill-conditioned). The full step-refusal path never once fired — a single stride halving always produced an acceptable candidate (backtrack rates 47% well-conditioned, 80% ill-conditioned at $\eta=0.1$).

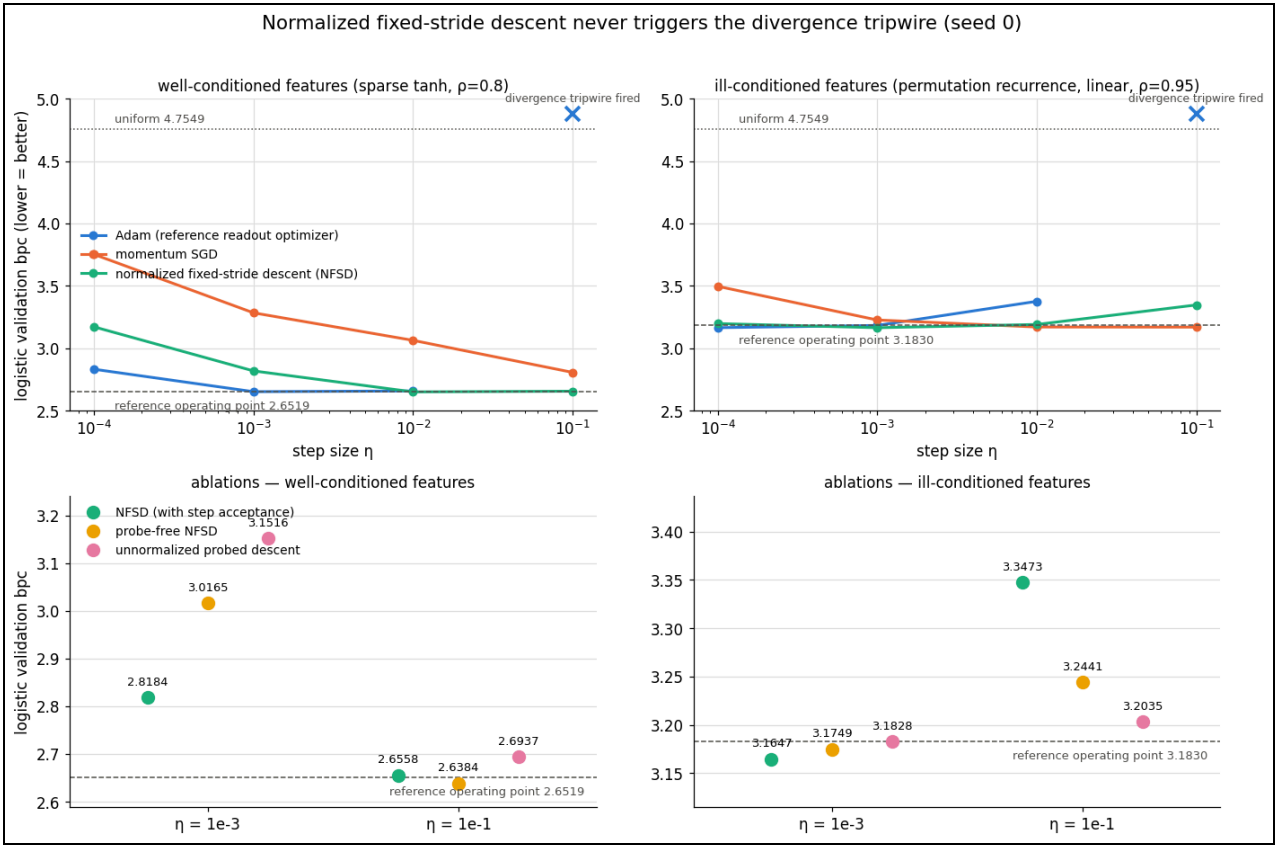


Figure 1. Seed 0 summary (internal run log #8). Top row: logistic validation bpc vs step size η (log scale) for the reference Adam configuration, momentum SGD, and NFSD on the well-conditioned (left) and ill-conditioned (right) feature sets; $\times$ marks a divergence-tripwire event (reference Adam at $\eta=0.1$, both feature sets); dashed lines: the reference instrument anchors 2.6519 / 3.1830; dotted: the uniform-prediction baseline 4.755. Bottom row: ablations at $\eta = 1e-3$ and $1e-1$ — full NFSD vs noprobe vs nonorm; the well-conditioned noprobe@0.1 (2.638) is the best cell in the grid, below the reference anchor.

Table 2. The registered replication, seed 1 (internal run log #9): 4/4 ordinals, 2/2 bands.

claim	seed 0	seed 1
R1 ill-cond.: best NFSD vs instrument setting	3.1647 vs 3.1830 (Δ 0.0183)	3.1651 vs 3.1823 (Δ 0.0172)
R2 well-cond.: noprobe@0.1 vs instrument setting	2.6384 vs 2.6519 (Δ 0.0135)	2.6325 vs 2.6467 (Δ 0.0142)
R3 step probing harmful at $\eta=0.1$ (both feature sets)	+0.0174 well-cond. · +0.1032 ill-cond.	+0.0038 well-cond. (thin) · +0.095 ill-cond.
R4 NFSD voids in sweep	0/8; adam@0.1 voided $\times 2$	0/8; adam@0.1 voided (well-cond.), 8.5978 bpc unvoided (ill-cond.)

Two disclosures from the rerun, recorded as registered. The well-conditioned margin in R3 (0.0038) is thin — inside the tie band other claims in this program use; it was judged as registered (a strict inequality) and flagged in the run journal. Second, seed 1 produced a new single-seed observation, *not* promoted: on the ill-conditioned set, Adam at $\eta=0.1$ diverged to 8.5978 bpc — nearly double the uniform-prediction baseline (4.7549) — without ever tripping the divergence tripwire, because the loss oscillated rather than rising monotonically and no single epoch's increase crossed the threshold. The

tripwire detects monotone blow-up, not sustained oscillation. Across both seeds, the reference Adam configuration at $\eta=0.1$ therefore voided in 3/4 cells, with unvoided divergence in the 4th. NFSD: 0 voids in all 16 sweep cells.

§5 Discussion

First, the instrument finding, which is the principal result. The program's frozen readout instrument is mis-calibrated in a conditioning-dependent way: its fixed learning rate is off-optimum by 0.017–0.018 bits on the ill-conditioned feature set (recovered by adam@1e-4 or a normalized fixed stride at 1e-3) and by 0.013–0.014 bits on the well-conditioned feature set (best cell in the grid: normalized fixed stride 0.1). The program action, entered in the program's confirmed-findings record, is that cross-feature-set comparisons of logistic *levels* carry a ~ 0.02 -bit learning-rate $\times$ conditioning error bar. The instrument nevertheless stays frozen — comparability across the existing archive outranks 0.02 bits — and separate evidence (internal run log #5) indicates that matched-parametrization *gaps* remain robust even where levels are biased.

Second, the optimizer. Normalized fixed-stride descent with exponentially weighted moving-average momentum needs no per-parameter state, never voided across four decades of step size on two seeds and two feature sets, and either ties tuned Adam to within 0.001 or beats it. Its stability derives entirely from the normalization: the step length is fixed regardless of gradient magnitude.

Third, the attribution. Minibatch step acceptance — the component the design brief made most distinctive — was strictly harmful at large stride, in all four cells and on both seeds. The safety of the method comes from normalization, not from step probing. The original run report's summary stands in substance: several registered ordinal predictions missed onto their pre-named alternatives, the quantitative bands held, and the step-acceptance component proved dispensable.

§6 References

1. D. P. Kingma, J. Ba, "Adam: a method for stochastic optimization," *ICLR* 2015 (arXiv:1412.6980).
2. B. T. Polyak, "Some methods of speeding up the convergence of iteration methods," *USSR Computational Mathematics and Mathematical Physics* 4(5), 1964.
3. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
4. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

§7 Provenance

- **32caa262** — 2026-08-11-caity-goats-on-a-mountain: the original design brief and main grid, 32 fits, seed 0 (internal run log #8). Registered before running; the pilot run was disclosed in the registration, including a known pilot-scale inversion artifact against which the predictions were held.
- **8a0a29b4** — 2026-08-11-goat-descent-rerun1: full 32-fit grid on seed 1, fresh reservoir draws and batch order (run ledger #9). Promotion by the pre-stated R1+R2 rule on two independent

passes.

All claims are judged strictly against the registered wording, with misses included above; 2 seeds $\times$ 2 feature sets; replicated before publication. Internal designation: goat descent.

AuxiLab · findings are pre-registered and replicated before publication

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